

Guanjin Li

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EDUCATION

Northwestern University, McCormick School of Engineering

Evanston, IL

Master of Science in Electrical Engineering (Thesis Track)

Sep. 2025 – Dec. 2026 (expected)

Cumulative GPA: 3.744/4.0

University of California San Diego, Revelle College

San Diego, CA

Bachelor of Science in Electrical Engineering and Applied Mathematics (Double Major)

Sep. 2021 – Jun. 2025

Depth: Machine Learning and Controls

PUBLICATIONS

Li, G. "CSNet-PGNet: Algorithm for Scene Text Detection and Recognition." *2022 3rd International Conference on Computer Vision, Image and Deep Learning & International Conference on Computer Engineering and Applications*, 2022. DOI: 10.1109/CVIDLICCEA56201.2022.9824815.

Li, G. "Six-Degree-of-Freedom Pose Estimation of Class-Level Objects Based on P2T-Net." *Information Technology and Control*, 2024. DOI: 10.5755/j01.itc.53.3.36121.

RESEARCH EXPERIENCE

Researcher, Project: High-Level Human-Robot Interaction, Northwestern's Center for Robotics and Biosystems

Feb. 2026 – Present

Advisor: Prof. Randy Freeman | Northwestern's Department of Electrical and Computer Engineering

- Focused on humanoid manipulation and physical human-robot interaction, developing collaborative frameworks between humans and humanoid robots for complex tasks with expectation modeling and motion prediction;
- Developed mathematical formulas and analyzed theoretical conditions to support human-humanoid collaboration, building upon the lab's prior project, MoCoBots;
- Designed experiments and simulated humanoid behaviors in virtual environments, followed by debugging simulation and deployment issues and deploying verified algorithms onto real humanoid robots;
- Tested the robustness of deployment and integrated individual deployment modules to achieve overall objectives;
- Independently led the entire research project, from outline construction, theoretical formulation and experiment design to real-robot implementation and validation.

Research Assistant in College of Computer Science, Nankai University

Jul. 2025 – Sep. 2025

Advisor: Prof. Jian Han | College of Computer Science, Nankai University

- Researched computational performance evaluation methods for automotive AI chips in intelligent driving systems;
- Compiled representative driving scenarios, chip-performance indicators, benchmark models, and datasets;
- Conducted model compatibility, frame rate, and latency tests; analyzed experimental data to identify performance bottlenecks;
- Surveyed 29 industry organizations, including vehicle manufacturers, component suppliers, algorithm providers, AI chip vendors, and testing institutions;
- Configured batch parameters, model sizes, test threads, and optimization modes to evaluate hardware-platform performance.

Researcher, Individual Project: Six-Degree-of-Freedom Pose Estimation of Class-Level Objects Based on P2T-Net

Jan. 2023 – Jun. 2024

Advisor: Prof. Shanjiang Tang | College of Intelligence and Computing, Tianjin University

- Thoroughly examined the 6D pose estimation algorithm based on deep learning, and proposed a class-level 6-DOF attitude estimation network featuring a simple concept and high data efficiency to enhance the accuracy of image classification, semantic segmentation, object detection, and instance segmentation under resource-constrained hardware settings;

- Introduced the implicit deep recovery technology in the cross-mode fusion stage to boost the RGB-D feature representation ability, employed the compact pyramid refinement operation to fuse multilayer features rapidly and effectively with fewer parameters;
- Completed a research paper which has been published in the SCI-indexed journal *Information Technology and Control*.

Researcher, Group Project: CSNet-PGNet: Algorithm for Scene Text Detection and Recognition

Jan. 2022 – Jun. 2022

Advisor: Prof. Rabih Younes | Dept. of Electrical & Computer Engineering, Duke University

- Explored artificial intelligence and computer vision in great depth, including machine learning, deep learning, object detection, reinforcement learning, and convolutional neural networks;
- Performed research on automatic recognition method of scene text, conducted experiments, collected and analyzed data, and contributed to the completion and publication of the paper;
- The integrated interactive system of Scene Text Detection and Recognition V1.0 has obtained the computer software copyright from the National Copyright Administration of China.

Researcher, Project: Fully Automatic Facial Deformation Transfer, Tencent Rhinoceros Elite Talent Training Plan

Jan. 2021 – Dec. 2021

Advisor: Prof. Lin Gao | PhD supervisor, University of Chinese Academy of Sciences

- Surveyed and analyzed emerging trends in AI, graphics, big data, network security, and determined academic direction;
- Established a thesis topic, prepared literature review, designed test plans, and strengthened technical foundations in AI and computer graphics;
- Related papers won the second prize (Top 3%).

Researcher, Project: Correction of Visual Distortion, China Association for Science and Technology

Jan. 2019 – Dec. 2019

2019 “Yingcaijihua” Science Talent Program, selected as 1 of the 6 students in Tianjin to attend the program

Advisor: Prof. Ming-ming Cheng | College of Computer Science, Nankai University

- Developed a scanning program to correct image distortion, and completed the project *Application of Adaptive Image Scaling Method Based on Hybrid Energy in the Correction of Visual Distortion of Curved Documents*;
- Won the First Prize in Tianjin Youth Science and Technology Innovation Competition and won the Certificate of Excellence out of 100 participants in the final competition.

COURSE PROJECTS

Autonomous Vehicle Perception Failures and Mitigation Strategies

Evanston, IL

Instructor: Prof. Qi Zhu | CE 495: Connected and Autonomous Vehicles: Challenges and Design | Winter 2026

- Analyzed perception and planning failures in autonomous vehicle corner cases;
- Compared camera-only and multi-modal AV perception systems using recent literature on robustness, sensor fusion, and open-world detection;
- Proposed mitigation strategies including sensor fusion improvements, conservative planning, audio-based emergency vehicle detection, and V2X/V2I communication, and simulated the proposed strategies using CARLA.

Object Detection and Avoidance for an Autonomous Vehicle

San Diego, CA

GitHub: <https://github.com/ZTKZTK13/summer-2024-final-project-team-3.git>

Instructor: Prof. Jack Silberman | ECE 148: Introduction to Autonomous Vehicles | Summer 2024

- Built and configured the Robotic Vehicle with Jetson Nano, VESC, OAK-D camera, GNSS, and LiDAR;
- Collected data and trained the object detection AI using Roboflow;
- Implemented object detection and avoidance through ROS 2 package coding and PID control;
- Synthesized the object detection and avoidance package with line tracking package and GNSS navigation.

Autonomous Driving Line-Tracking Chassis

San Diego, CA

Instructor: Prof. Karcher Morris | ECE 5: Introduction to ECE | Winter 2023

- Used Arduino to design and implement the PID control schematic level simulations and coding;
- Designed and implemented the chassis model by 3D printing, and synthesized the motor and circuit by soldering;
- Adjusted the parameter of the PID control and the photoresistors to optimize the performance in edge cases.

INTERNSHIP EXPERIENCE

Tianjin Ruipeng Medical Equipment Co., Ltd.

Tianjin, China

Research Assistant in Exoskeleton Robot Laboratory

Jul. 2023 – Sep. 2023

- Assisted researchers to enhance the accuracy of human motion intention recognition and improve human-machine collaborative control technology;
- Collected experimental data, summarized and analyzed results, proposed improvement suggestions, and wrote and modified programs.

Tianjin Yubang Biological Technology Development Co., Ltd.

Tianjin, China

Test Engineer in R&D Department

Jun. 2022 – Sep. 2022

- Participated in the testing of the new generation of food additives and solid drinks in the company;
- Assisted researchers in understanding product design principles, developing test plans and schemes, writing and executing software tests, and analyzing test data;
- Investigated defects and deficiencies in the tests to evaluate the best solution to the problem;
- Worked with the developers to improve the feasibility and reliability of the solution.

TRAINING EXPERIENCE

Rhino Bird Open Source Talent Training Program – “Advanced Study Series Courses”

Jun. – Sep. 2022/2023/2024

Tencent Computer Systems Co., Ltd., Shenzhen, China | Online Research

- Mastered the general knowledge of open source and the key technical points of the project;
- Gained experience in open-source programming through two modes: single challenge, multiplayer competition;
- Conducted research on the open-source project “NCNN – Neural Network Inference Framework”.

SKILLS & INTERESTS

- **Software and Programming Skills:** Proficient in C, C++, Java, Python, PyTorch, MATLAB, ROS 2, R, PSpice, Verilog
- **Strengths:** Machine Learning, AI, Control Systems Design, Robotics Design, Mathematics
- **Interests:** Playing piano, Reading, Hiking, Swimming, Playing basketball, Surfing
- **Language Skills:** English (Proficient); Chinese (Native); Italian (Elementary)